

COS 214 PROJECT

PLANT
NURSERY



BRAINS BEHIND THE PROJECT

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FUNCTIONAL REQUIREMENTS

1. Greenhouse Management
2. Staff Management & Actions
3. Customers & Sales Management



GREENHOUSE MANAGEMENT

PLANT CARE HANDLING

The system manages the care routines for the different plant types, such as management of water amounts, fertilization, and other needs. It is different for every plant type.

PLANT LIFE CYCLE

The system manages the plants from planting seeds to growth to selling of plants. Selling of plants are determined by how far along each plant has grown.

INVENTORY TRACKING

The system maintains an inventory system that keeps track of all plants. Updating the inventory happens when plants have grown, been sold, or require care.

STOCK ADDITION

The system allows for adding new plant stock or plant types to the nursery.



STAFF MANAGEMENT & ACTIONS

PLANT TENDING

Staff does care routines on plants based on plant type.

INVENTORY COORDINATION

Staff keeps track and updates the inventory making sure that the greenhouse and sales are in sync.

CUSTOMER ASSISTANCE

Staff interacts with the customers by showing customer what is available for them to purchase.

ROLE EXTENSIBILITY

The system supports extending roles of staff (eg: delivery staff).



CUSTOMERS & SALES MANAGEMENT

CUSTOMER BROWSING

Customers can view available plants on sale based off of the inventory.

STAFF INTERACTION

Customers receive guidance from staff (available plants), purchasing from the staff, and change the inventory when a purchase is made.

SALES & TRANSACTIONS

The system handles plant purchases, including updating the inventory in real-time and processing the transactions.

CUSTOMER DECISION IMPACT

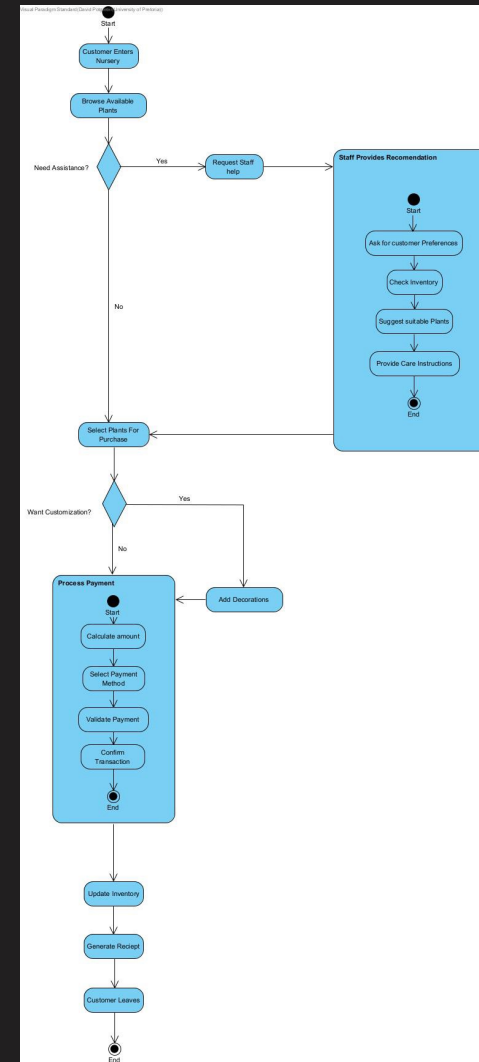
Customer actions (eg: buying a plant) affects the nursery operations, such as decreasing the stock of the inventory.



UML DIAGRAMS

1. State Machine
2. Sequence
3. Activity
4. Object
5. Communication
6. Class

ACTIVITY DIAGRAM

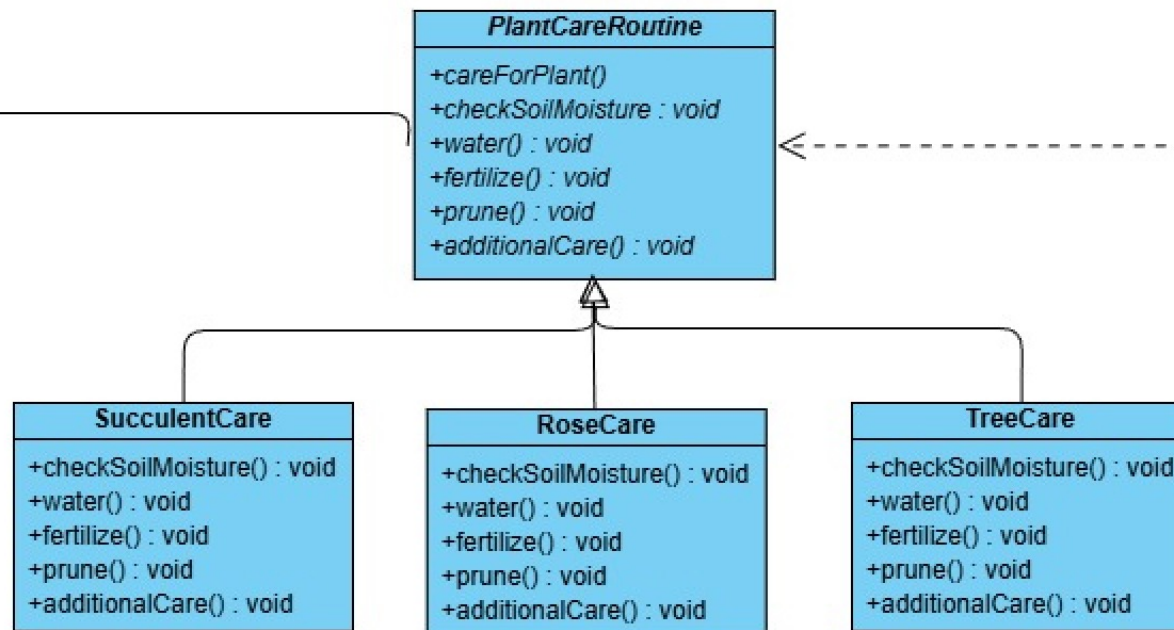


CLASS DIAGRAMS

1. Template Method
2. Iterator
3. Adapter
4. Composite
5. Prototype
6. Builder
7. State
8. Strategy
9. Factory Method
10. Decorator
11. Command

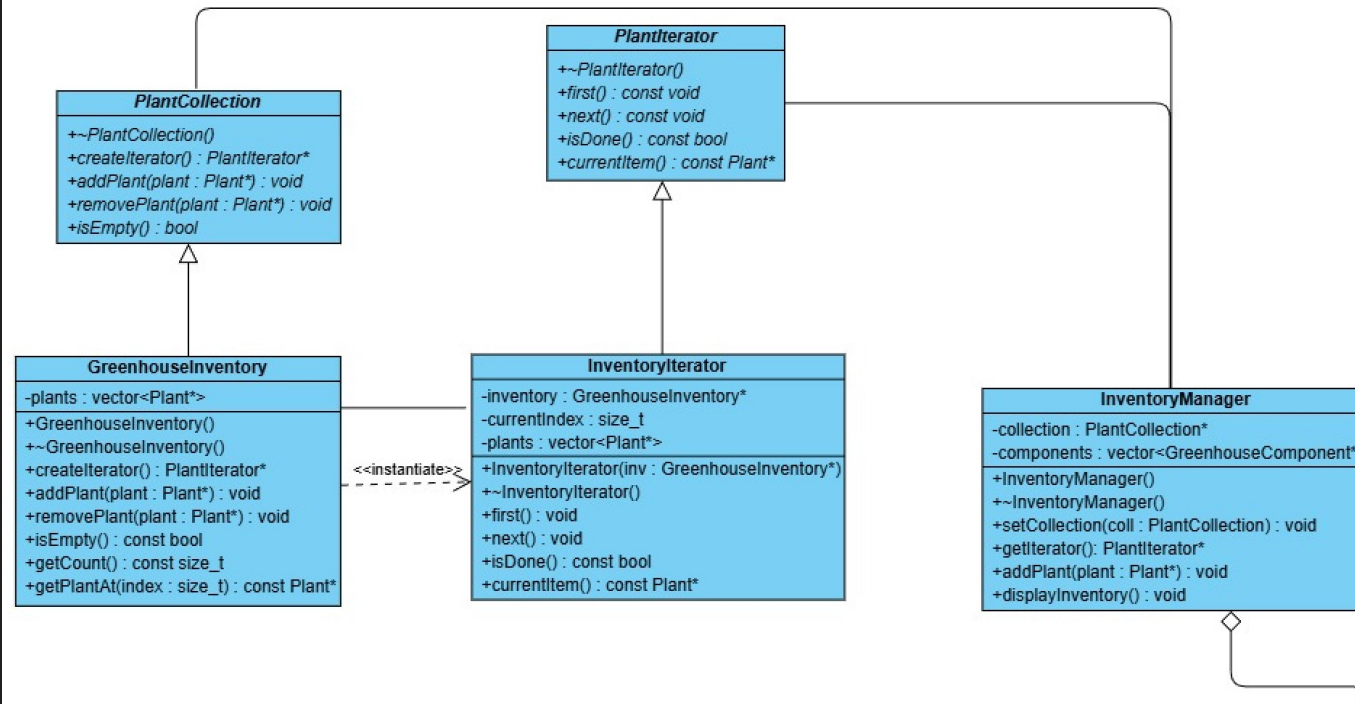
TEMPLATE METHOD

Template:
Define care routine skeleton, let
subclasses customize steps



ITERATOR

Iterator: Traverse inventory without exposing its structure

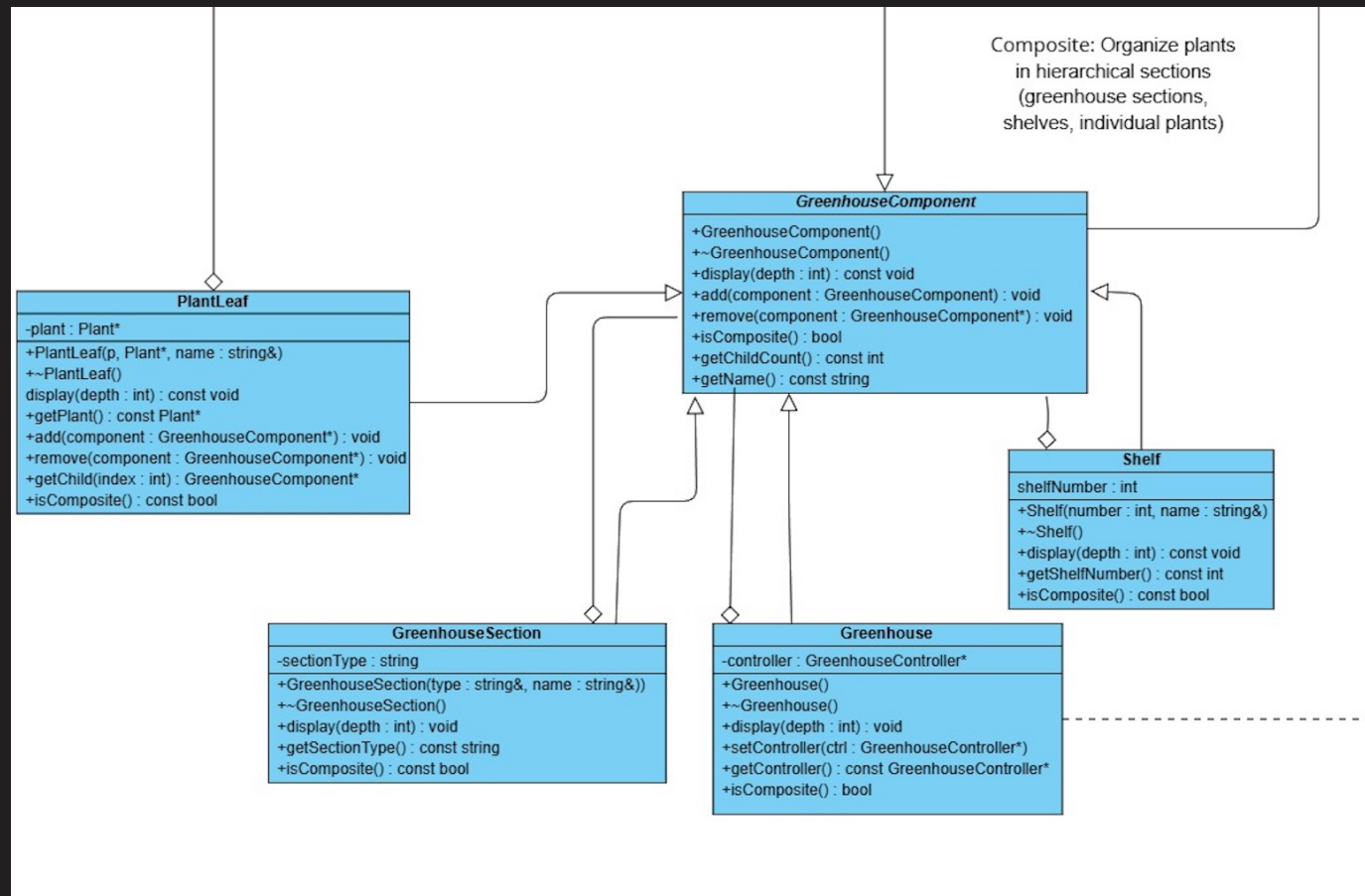


ADAPTER

Adapter - Legacy Watering System

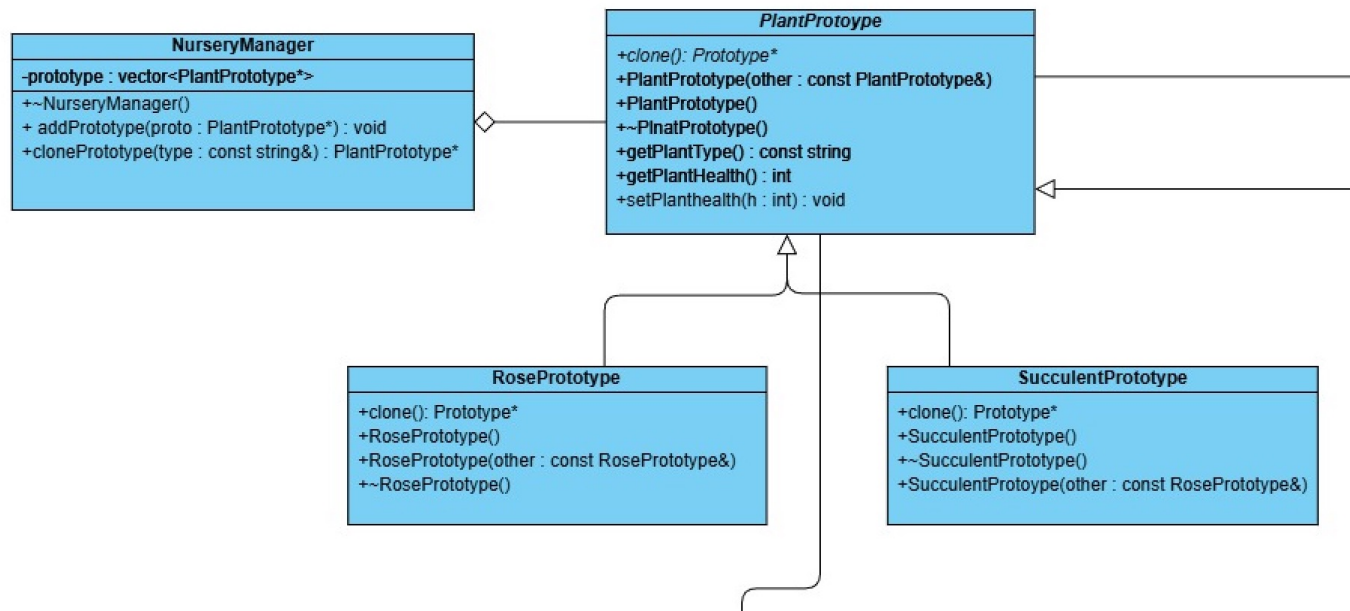


COMPOSITE



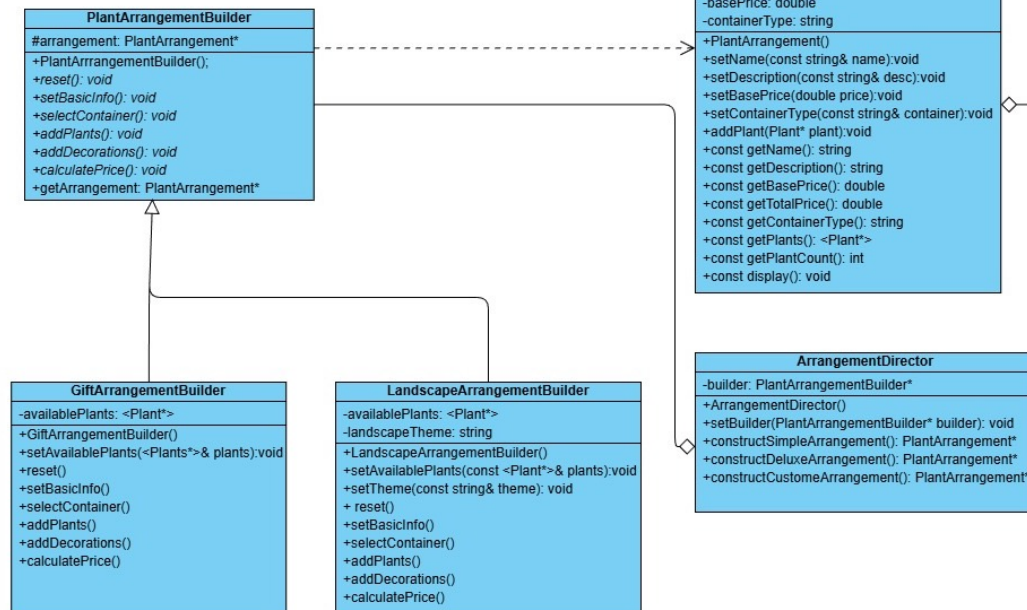
PROTOTYPE

Prototype used to make a clone of a plant
for propagation

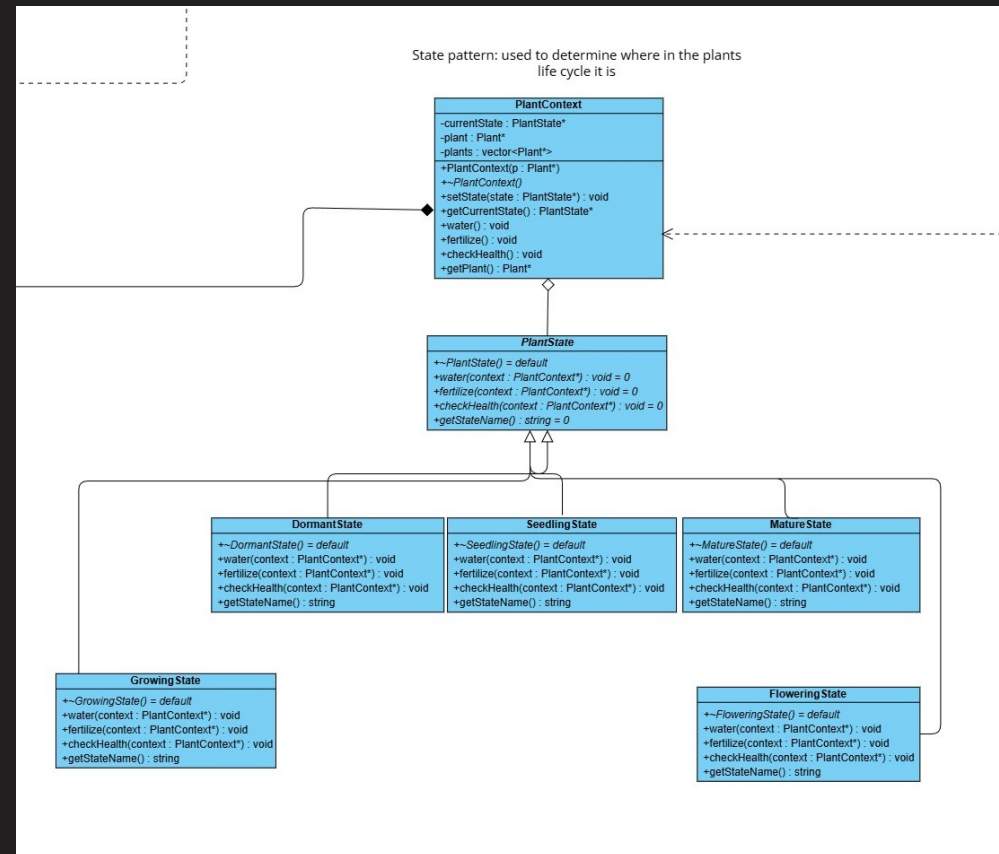


BUILDER

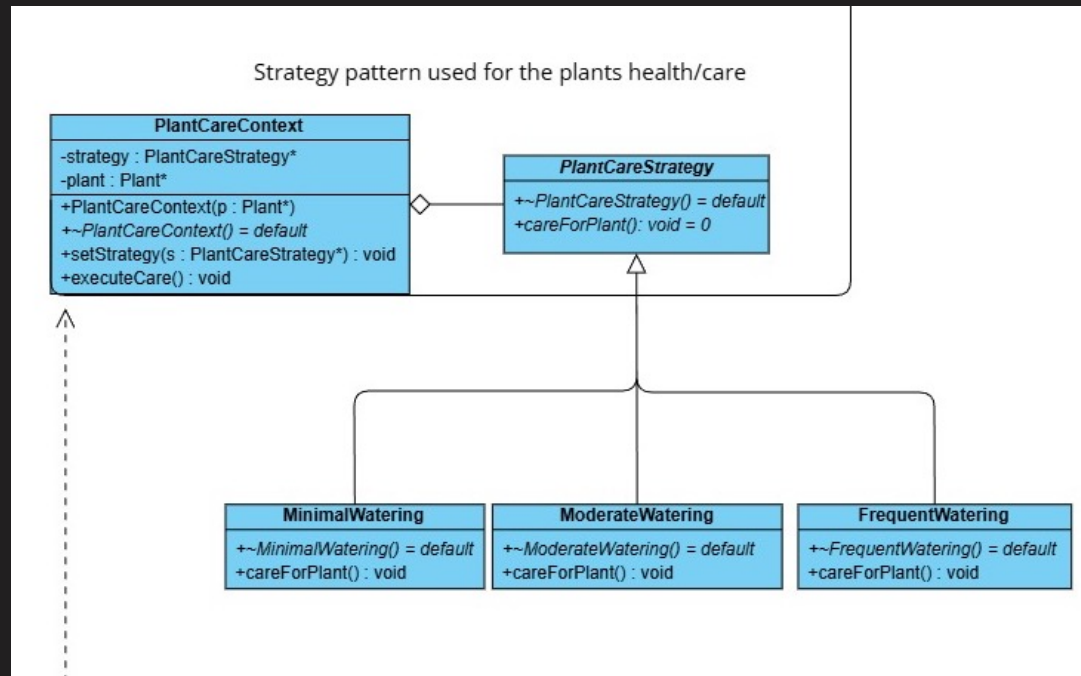
A builder to create complex plant arrangements step by step



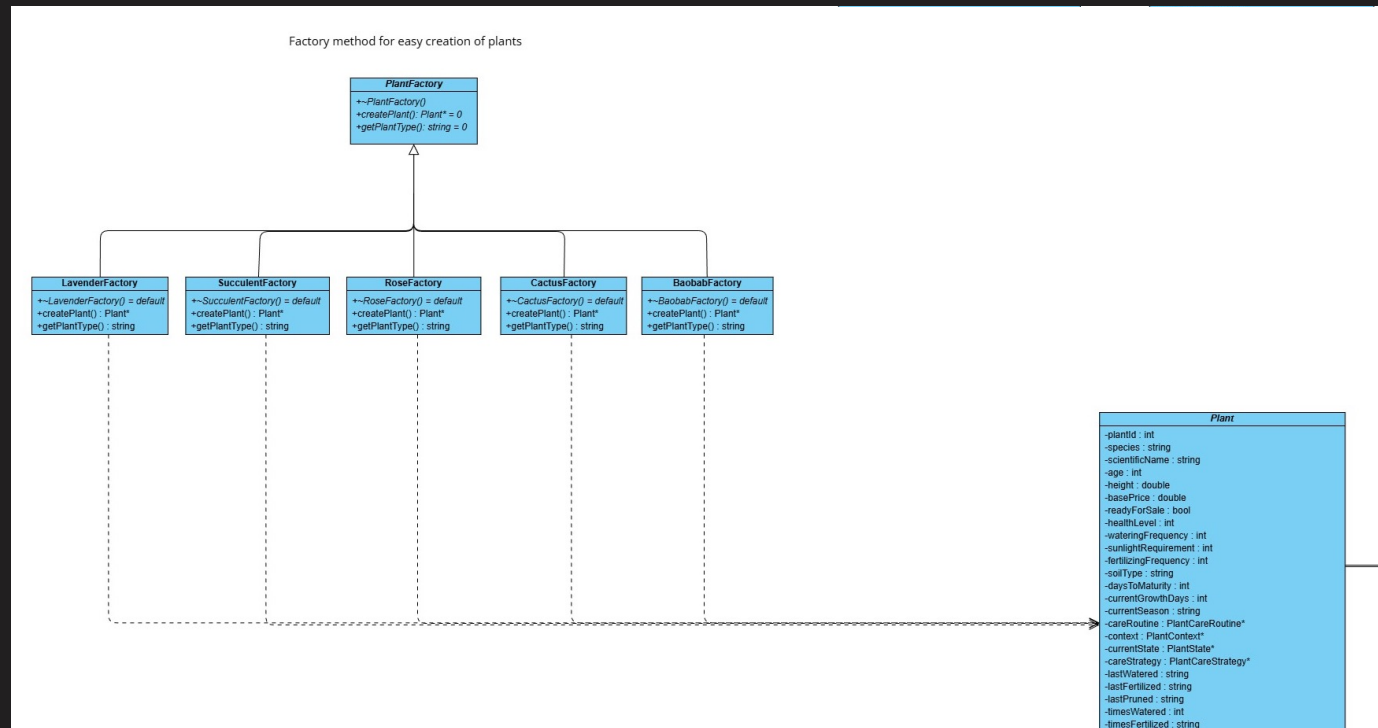
STATE



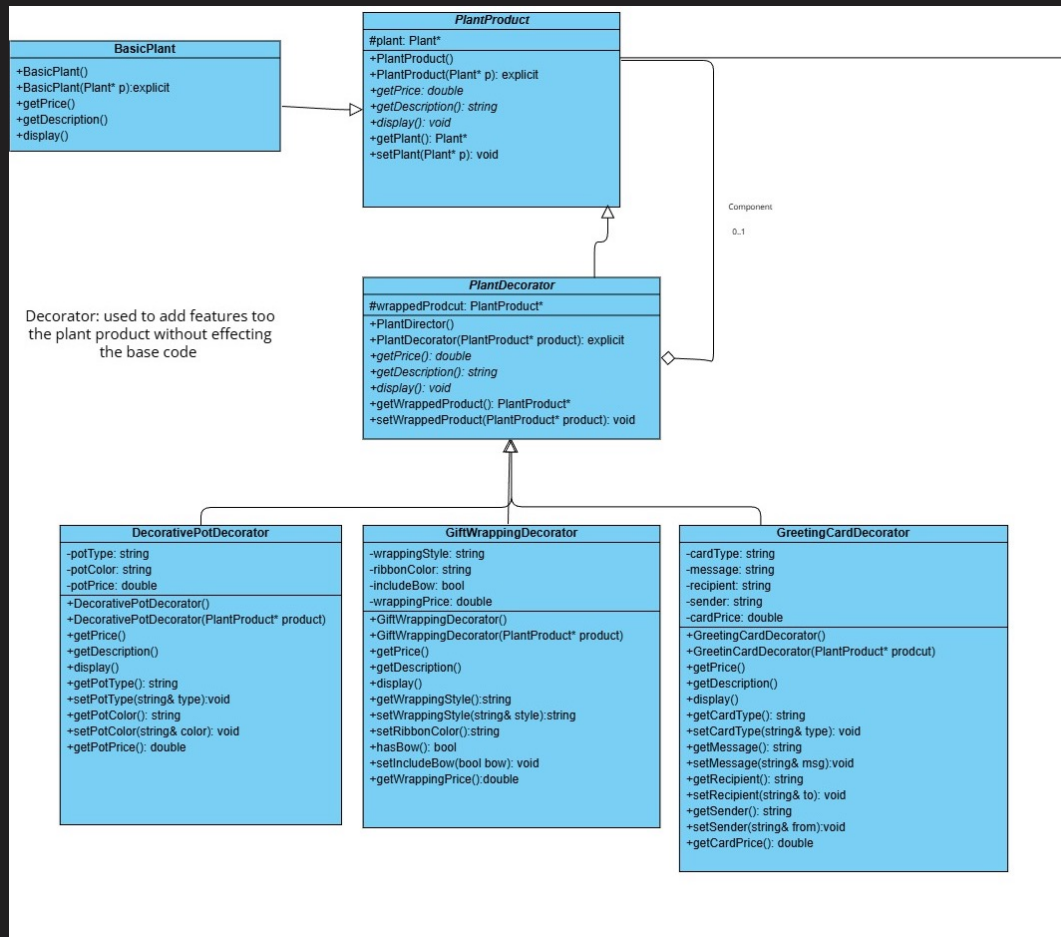
STRATEGY



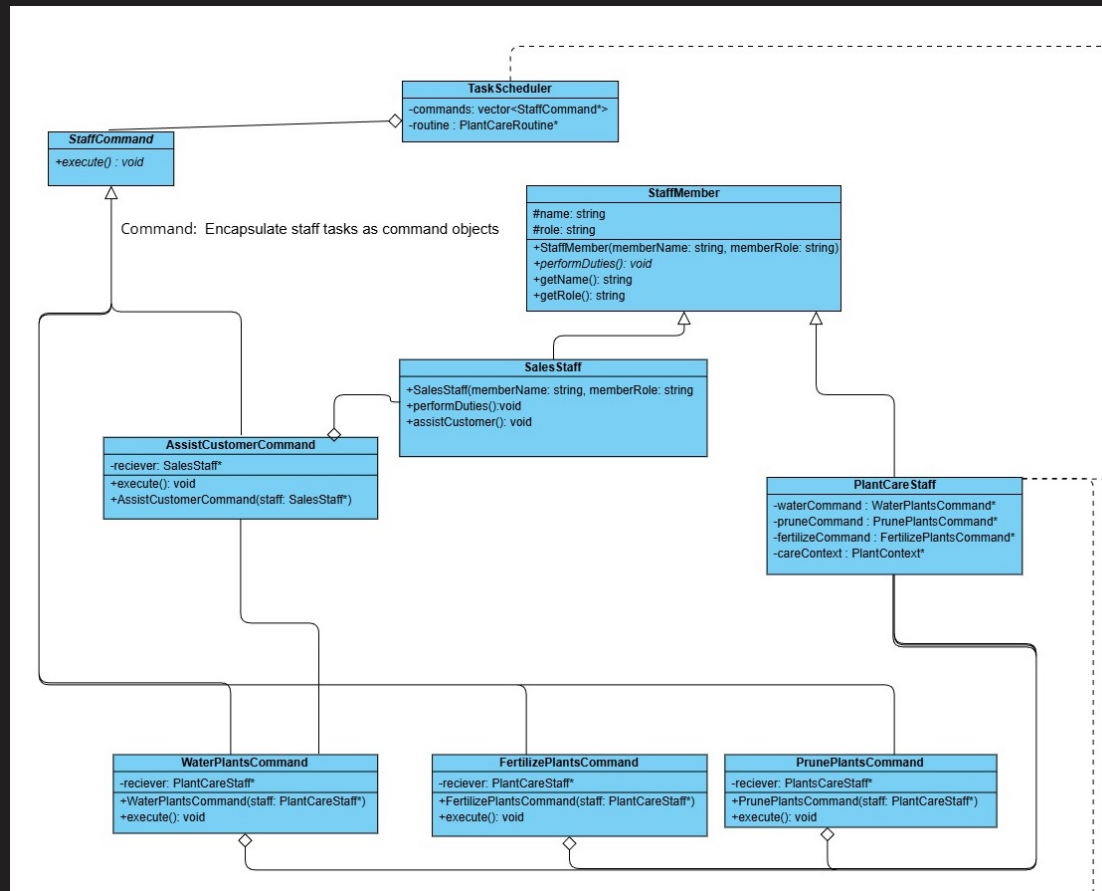
FACTORY METHOD



DECORATOR



COMMAND



GITHUB

Commits



GITHUB

Branches

Branches

[New branch](#)

Overview Yours Active Stale All

Q Search branches...

Default

Branch	Updated	Check status	Behind	Ahead	Pull request
<code>main</code>	5 minutes ago	✓ 3 / 3			Default ...

Your branches

Branch	Updated	Check status	Behind	Ahead	Pull request
<code>dev</code>	40 minutes ago		14	0	#8 ...

Active branches

Branch	Updated	Check status	Behind	Ahead	Pull request
<code>dev</code>	40 minutes ago		14	0	#8 ...
<code>GUI-Testing</code>	2 hours ago		39	6	...

GITHUB

Pull Requests

Label issues and pull requests for new contributors

Dismiss

Now, GitHub will help potential first-time contributors [discover issues](#) labeled with [good first issue](#)

Filters

is:pr is:closed

Labels 9

Milestones 0

New pull request

Clear current search query, filters, and sorts

0 Open

8 Closed

Author

Label

Projects

Milestones

Reviews

Assignee

Sort

Dev

#8 by McRobbie24646866 was merged 32 minutes ago

Moving the dev into main

#7 by DillonKoekemoer was merged yesterday

Dev

#6 by Noah-Dollenberg was merged 3 days ago

Implemented Adapter design pattern

#5 by Noah-Dollenberg was merged last week

Dev

#4 by Noah-Dollenberg was merged last week

Builder being added

#3 by DillonKoekemoer was merged last week

Joshua Done

#2 by JoshRoberts136 was merged 2 weeks ago

Dillon

#1 by Noah-Dollenberg was merged 2 weeks ago

**AND FOR
OUR NEXT
TRICK...**